

# RF and Microwave Engineering Exam Notes

EX 716 (BEI) / EX 752 (BEX) · IV-I / IV-II · TU, IOE · tiers over 24 papers, 2069–2082 BS

## 5. Microwave Generators

5 hrs · asked in **23 of 24 papers** · 5 syllabus sub-topics · **166 marks, 8.7% of the archive** → two tubes (klystron, magnetron) answer most of it; one idea (bunching) is the 2-mark rider on almost every question.

### READ THIS FIRST

Chapter 5 is descriptive, with no numerical PYQ in 24 papers, and the questions come in a fixed shape: **a 2-mark definition, then a 6 to 8 mark “construction and working”**.

1. **The 2-mark riders** (bunching, density modulation, transit time, cross-field effect, slow-wave structure) → each has a boxed two-line answer below. Bunching alone is asked in 9 papers.
2. **Klystrons** (two-cavity, multicavity, reflex; 9 papers) → §5.3 and §5.4. One figure (the Applegate diagram) explains all three.
3. **Magnetron** (9 papers) → §5.5. The “45° / 60° / 90° phase shift between adjacent cavities” versions are the same answer plus one formula,  $\phi = 2\pi n/N$ .

Two things worth knowing before the paper starts:

- Every tube here does the same three things: **velocity-modulate** a beam, let it **bunch**, and **decelerate the bunches** in a field that takes their energy. Say this in the first line of any tube answer and the structure of the answer follows.
- Linear-beam (O-type) tubes: klystron, reflex klystron, TWT, BWO. Crossed-field (M-type) tubes: magnetron, crossed-field amplifier. An examiner who writes “field effect” or “cross-field” wants the M-type family.

### 5.1 Conventional Tubes at Microwaves and the Transit-Time Effect

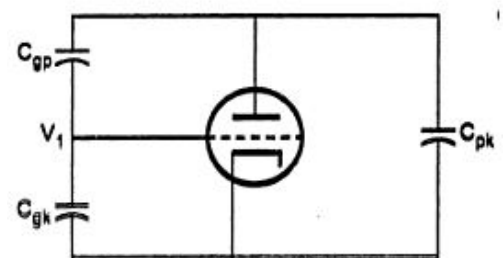
**How is the output of conventional tubes reduced at microwaves? What is transit-time effect?** **2 2+6**

**2+8**

**HOT 3/24** 70 Ma, 77 Ch, 79 Bh · always the 2-mark opener to a tube question

**Why triodes and pentodes fail above about 1 GHz**

- **Inter-electrode capacitance**  $C_{gp}, C_{gk}, C_{pk}$ :
  - Reactance  $1/\omega C$  falls with frequency → shorts the input and output → gain falls.
  - Cure (spread electrodes, smaller area) raises transit time: a **compromise**.
- **Lead inductance**:  $\omega L$  grows → degenerative feedback; cure: short leads (disk-seal tubes).
- **Transit-time effect** (below).
- **Cathode emission**: more current needs a bigger cathode, which adds capacitance; filament voltage and temperature are limited.
- **Power losses**: skin-effect resistance grows as  $\sqrt{f}$ ; dielectric loss in the glass envelope; radiation from leads.



Inter-electrode capacitances of a triode. Karkee, Ch-5 deck, slide 4.

**Transit-time effect: the 2-mark answer** 2

- **Transit time** = time an electron takes from cathode to anode. **Transit-time effect**: when this becomes a significant fraction of one RF cycle, the grid voltage changes while the electron is in flight → plate current lags the grid voltage → efficiency and gain collapse.
- Numbers:  $\tau = 1$  ns is 0.001 cycle at 1 MHz but a **whole cycle** at 1 GHz. Above about 0.1 cycle the loss is serious.
- Estimate: 100 V gives  $v = \sqrt{2eV/m} = 5.93 \times 10^6$  m/s; a 1 mm gap then takes  $3.4 \times 10^{-10}$  s, i.e. 0.34 cycle at 1 GHz.
- Cure in a conventional tube: closer electrodes and higher voltage (fights the capacitance). **Cure in a microwave tube: exploit it.** Klystrons, TWTs and magnetrons turn transit time into velocity modulation and bunching.

**5.2 Velocity Modulation, Density Modulation and Bunching****What is bunching effect (density modulation)? Explain velocity modulation and bunching** 2 5 10

**TOP 9/24** 70 Bh, 71 Bh, 72 Ash, 72 Ma, 73 Ma, 75 Bh, 80 Ba, 80 Bh, 81 Bh · **the most repeated ask in the chapter**

**2-mark answer. Velocity modulation**: an RF voltage across a gap speeds up electrons crossing in one half-cycle and slows those crossing in the other. **Bunching (= density modulation)**: in the field-free drift space beyond, fast electrons catch up with slow ones sent earlier, so the uniform beam gathers into **bunches** once per RF cycle. The beam current is now modulated at the RF frequency and can drive a second cavity.

**The equations (for the 5 and 10 mark versions)**

- Beam accelerated by  $V_0$ :  $v_0 = \sqrt{2eV_0/m} = 0.593 \times 10^6 \sqrt{V_0}$  m/s.
- Buncher gap voltage  $V_1 \sin \omega t$ . Leaving at  $t_1$ :

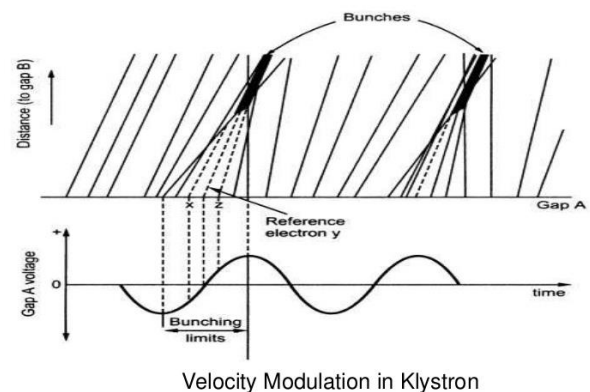
$$v(t_1) = v_0 \left[ 1 + \frac{\beta_i V_1}{2V_0} \sin \omega t_1 \right]$$

$\beta_i = \frac{\sin(\theta_g/2)}{\theta_g/2}$ , the **beam coupling coefficient**;  $\theta_g = \omega d/v_0$  the gap transit angle.

- Three reference electrons: #2 crosses at zero voltage (unchanged), #3 on the rising half (accelerated), #1 before it (decelerated). #1 and #3 converge on #2.
- Arrival at the catcher, drift length  $L$ :  $t_2 = t_1 + L/v(t_1)$ .
- **Bunching parameter**

$$X = \frac{\beta_i V_1}{2V_0} \theta_0, \quad \theta_0 = \frac{\omega L}{v_0}$$

- Catcher current:  $i_2 = I_0 + \sum_n 2I_0 J_n(nX) \cos n\omega(t_2 - \tau_0)$ .
- Fundamental  $2I_0 J_1(X)$  peaks at  **$X = 1.841$** , where it is  $1.164 I_0$  → optimum drift length.

**Two cavity klystron**

Applegate diagram: each line is one electron's distance vs time; steep lines are fast. Lines converge (a bunch) around the electron that crossed at zero gap voltage. Karkee deck, PDF p22.

- **Checked by simulation** in `tubes.py`: electrons launched with this velocity law and drifted ballistically give a fundamental of exactly  $2J_1(X)$ .
- Too short a drift ( $X < 1.84$ ): weak bunches. Too long: electrons overtake and the bunch spreads again (**debunching**).

### 5.3 Two-Cavity and Multicavity Klystrons

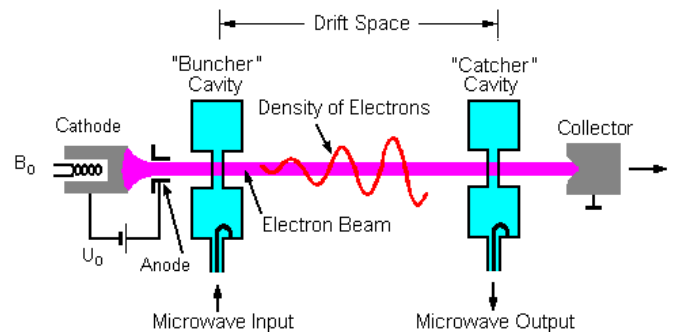
## Describe the construction and operation of a two-cavity / multicavity klystron amplifier or oscillator

4 5 2+7 2+8

TOP 7/24 69 Bh, 70 Ma, 71 Bh, 72 Ash, 78 Ch, 80 Ch, 82 Ba · 70 Ma two-cavity; the rest mostly multicavity

### Construction (two-cavity amplifier)

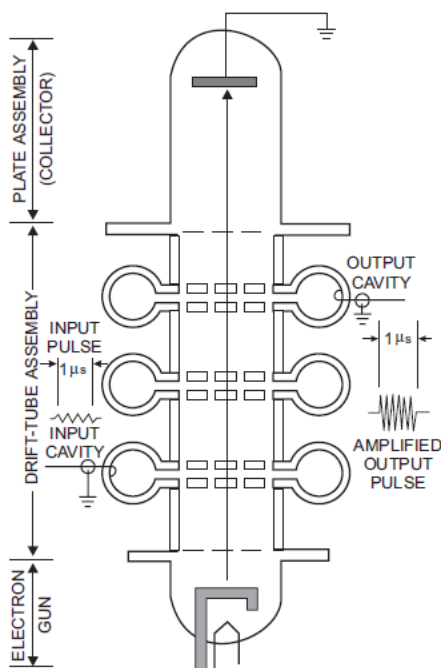
- **Electron gun:** heated cathode, focusing electrode, anode at  $+V_0$  forms a narrow beam; an axial magnetic field keeps it focused.
- **Buncher cavity (input):** re-entrant cavity with grids across the beam; RF in via coaxial loop.
- **Drift space:** field-free tube of length  $L$ .
- **Catcher cavity (output):** identical cavity, RF out via loop.
- **Collector:** absorbs the spent beam.
- Cavities and collector at the same dc potential, so the beam drifts at constant mean speed.



Two-cavity klystron. Karkee deck, slide 14.

### Operation

1. Weak RF in the buncher sets up  $V_1 \sin \omega t$  across its gap.
  2. **Velocity modulation:** electrons crossing are accelerated or decelerated (§5.2).
  3. **Bunching** in the drift space: velocity modulation becomes density modulation.
  4. Bunches reach the catcher once per cycle and **induce** RF current in it. Its field builds up to **oppose** the bunches: they are decelerated and give up kinetic energy, while the thin stream between bunches is accelerated and takes back only a little.
  5. Net energy flows from beam to RF  $\rightarrow$  amplified output. The beam leaves slower and is collected.
- **Correction to the source** (PDF p23): it says "all the kinetic energy is converted". Only part is; the beam still reaches the collector with most of its speed. Theoretical maximum efficiency is  $\frac{1}{2} \times 1.164 = 58\%$  (at  $X = 1.841$ , catcher voltage equal to  $V_0$ ); practical about 40%. verified/added
  - Gain of two cavities is low (about 10 dB)  $\rightarrow$  multicavity.



Multicavity klystron. Karkee deck, slide 19.

### Multicavity klystron (amplifier)

- One or more **intermediate cavities** between buncher and catcher, not externally driven.
- Weak bunches from cavity 1 excite cavity 2  $\rightarrow$  its larger gap voltage re-modulates the beam  $\rightarrow$  tighter bunches  $\rightarrow$  and so on.
- Each stage adds gain: **30 to 70 dB** overall.
- **Stagger tuning** the cavities slightly apart widens the bandwidth.
- Higher efficiency and power: up to MW pulsed, kW CW.
- Uses: UHF TV and troposcatter transmitters, satellite ground stations, radar transmitters, linear accelerators.

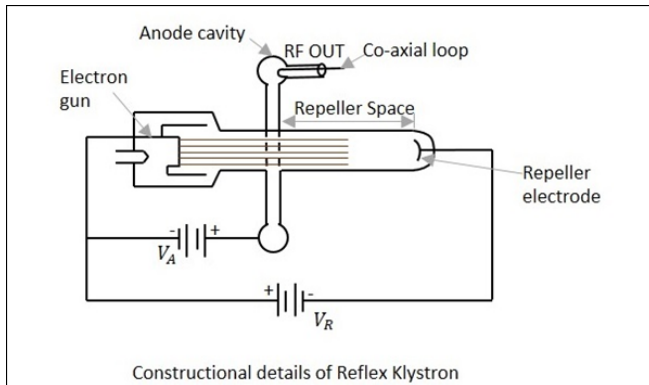
### Klystron oscillator

- Feed part of the catcher output **back** to the buncher with the right phase (feedback loop, slide 18)  $\rightarrow$  self-sustained oscillation. Multicavity oscillator: the same loop around a multicavity tube (72 Ash).
- Tuned by the cavities; for a single-cavity, voltage-tuned oscillator use the reflex klystron (§5.4).

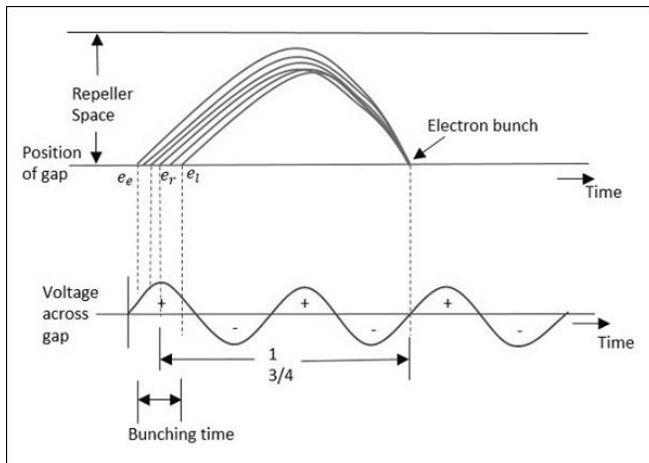
## 5.4 Reflex Klystron

Explain velocity modulation and bunching in a reflex klystron with neat diagrams and equations 5 10

PIN 2/24 72 Ma, 75 Bh · 75 Bh prints “multicavity reflex klystron”; a reflex klystron has one cavity



Karkee deck, PDF p29.



Electrons  $e_e, e_r, e_l$  return together: the bunch arrives  $n + \frac{3}{4}$  cycles later. PDF p30.

- **Electronic tuning:** changing  $V_R$  changes  $T'$  and pulls the frequency a few percent; modes are the different  $n$ .
- **Efficiency:** maximum  $\frac{2X'J_1(X')}{2\pi n + 3\pi/2}$  with  $X' = 2.405$ : 53% for  $n = 0$  (not practical), **22.7% for  $n = 1$** . The deck's 22.78% is this, with  $X'$  rounded to 2.408. Practical 10 to 20%.
- **Performance:** 4 to 200 GHz, 1 mW to 2.5 W. **Uses:** local oscillator in receivers, signal source for bench tests, portable links, pump for parametric amplifiers.

## 5.5 Cavity Magnetron and the Cross-Field Effect

Describe the construction and working of a cavity magnetron; magnetron with 45 / 60 / 90 degree phase shift between adjacent cavities 5 2+6 8 2+8 10

TOP 9/24 70 Bh, 71 Ma, 72 Ma, 74 Ma, 77 Ch, 79 Ch, 80 Ba, 81 Ba, 81 Bh · 81 Bh's "high-power oscillator using bunching and field effects" is this tube

**2-mark answer: cross-field effect** (72 Ma, 78 Ch, 81 Bh). An electron in **perpendicular** dc electric and magnetic fields feels  $\vec{F} = -e(\vec{E} + \vec{v} \times \vec{B})$ . It does not go straight to the anode: it follows a **cycloidal** path and drifts at  $v = E/B$ , perpendicular to both fields. Crossed-field (M-type) tubes use this drift to keep electrons in step with an RF wave for a long interaction.

### Construction

- Electron gun, **one** re-entrant anode cavity at  $+V_0$ , and a **repeller** electrode at  $-V_R$  beyond it.
- Output via a coaxial loop from the same cavity.

### Operation

1. Noise starts a small RF voltage across the cavity gap.
2. Electrons crossing it are **velocity-modulated**.
3. In the retarding repeller field they stop and turn back. A **faster** electron goes deeper and takes **longer**: round trip  

$$T' = \frac{2mLv}{e(V_R + V_0)}$$
4. So an early (fast) electron, the reference and a late (slow) one return **together**: the bunch forms in the **repeller space**.
5. The bunch re-crosses the gap. If the gap field then opposes it, it gives energy to the cavity and oscillation grows.

### Condition for oscillation

- The bunch forms round the electron that left at zero gap voltage (going positive), and must return when the field most strongly retards it:

$$T' = \left(n + \frac{3}{4}\right) \text{ cycles}, \quad n = 0, 1, 2$$

- Power returned to the gap  $\propto -J_1(X) \sin \theta_0$ : positive for  $\theta_0$  in the second half-cycle, largest at  $\frac{3}{4}$ . `tubes.py` scans  $\theta_0$  with simulated electrons and finds the peak at 1.750 cycles.

## Construction

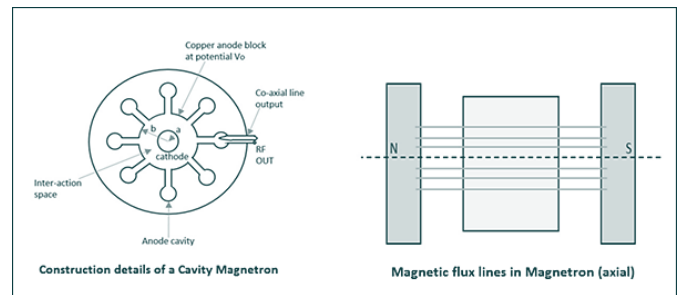
- Thick cylindrical **cathode** on the axis (radius  $a$ ).
- Copper **anode block** (radius  $b$ ) with  $N$  (typically 8) **resonant cavities** cut into it, opening onto the **interaction space**.
- Radial** dc field from anode voltage  $V_0$ ; **axial** magnetic field from a permanent magnet, so  $E \perp B$ .
- Output: coupling loop in one cavity to a coaxial line or waveguide.
- Straps** join alternate anode segments to hold the tube in the  $\pi$ -mode.

## Electron paths and the Hull cut-off

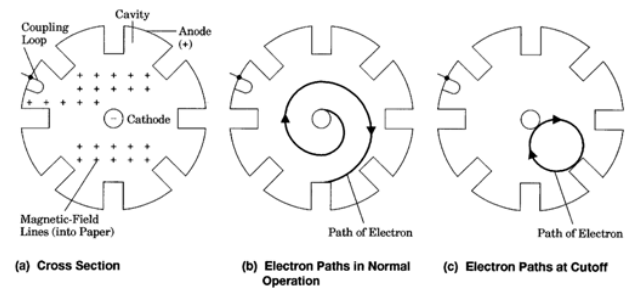
- $B = 0$ : straight radial path to the anode.
- Small  $B$ : curved path, still reaches the anode.
- $B = B_c$ : path just **grazes** the anode (cut-off).
- $B > B_c$ : electron returns to the cathode without collecting.
- Hull cut-off field**:

$$B_c = \frac{\sqrt{8mV_0/e}}{b(1 - a^2/b^2)}$$

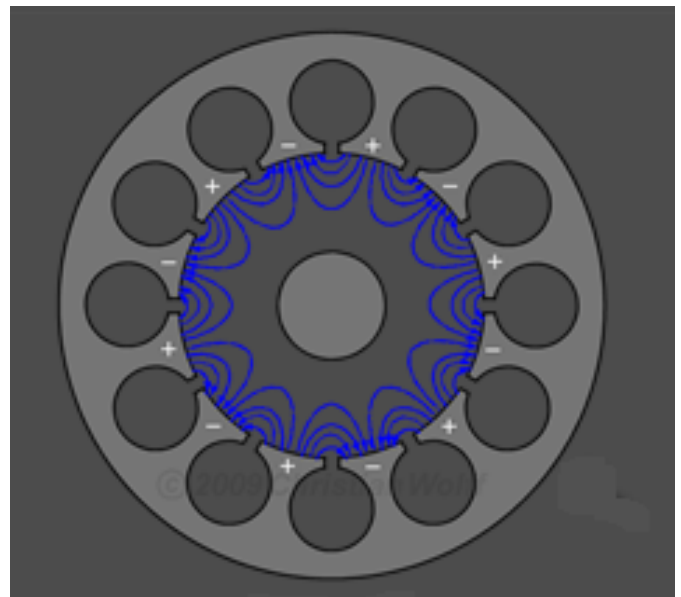
- Checked by integrating electron trajectories in `tubes.py`: for  $V_0 = 10$  kV,  $a = 5$  mm,  $b = 10$  mm the grazing field is 0.0899 T, equal to the formula. verified/added
- The magnetron works just **above**  $B_c$ : without RF no electron reaches the anode.



Cavity magnetron cross-section and axial flux. PDF p55.



(b) normal operation, (c) at cut-off. PDF p61.



$\pi$ -mode spoke pattern. Deck slide 50.

## Operation (oscillator)

- Noise excites the cavities; RF fringing fields appear at the vane tips, alternating round the anode.
- Electrons circling the cathode meet these fields. Those in a **retarding** RF field slow down, feel less  $v \times B$  force, and **drift outwards**, giving energy to the RF on the way.
- Those in an **accelerating** field curl back to the cathode (back-bombardment heats it, often enough to switch the heater off).
- The result is **phase focusing**: the electron cloud forms rotating **spokes**, each sitting in a retarding field. This is the magnetron's **bunching**.
- On the next half-cycle the field pattern reverses and the spokes rotate one cavity on, staying in step: sustained oscillation. Efficiency 40 to 70%.

## The phase-shift versions: 45, 60, 90 degrees between adjacent cavities 6+2 8 3+7

74 Ma, 77 Ch, 80 Ba (45°); 81 Ba (60°); 72 Ma (90°)

- Around a ring of  $N$  cavities the total phase must be a whole number of turns:

$$\phi = \frac{2\pi n}{N}, \quad n = 0, \pm 1, \pm 2 \text{ up to } \pm \frac{N}{2}$$

- $n = N/2$ :  $\phi = 180^\circ$ , the  $\pi$ -mode, normal operation.  
 $n = 0$ : zero mode, no RF between anode and cathode, unused.
- The RF pattern repeats every  $360^\circ/\phi$  cavities, so there is **one spoke** per that many cavities:
  - $45^\circ$ : 8 cavities per wavelength (in an 8-cavity tube  $n = 1$ , one spoke).
  - $60^\circ$ : 6 cavities per wavelength ( $N = 6$ ,  $n = 1$ ).
  - $90^\circ$ : 4 cavities per wavelength ( $N = 8$  gives  $n = 2$ , two spokes).
  - $180^\circ$ : 2 cavities,  $N/2$  spokes.

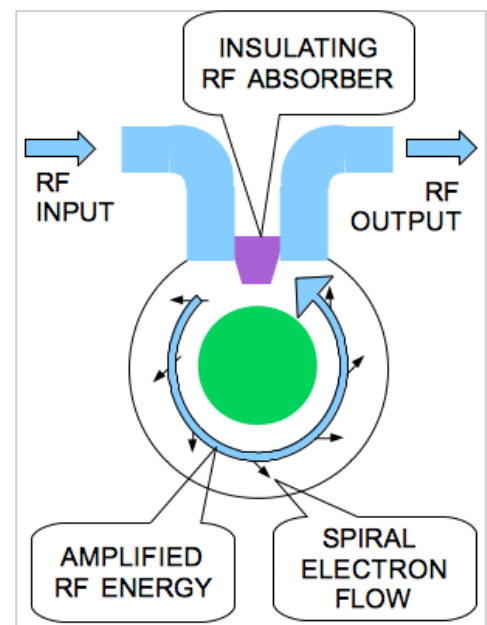
- **Synchronism**: the spokes rotate at the RF pattern's angular velocity  $\omega/n$ ; set  $V_0$  and  $B$  so the electron drift ( $E/B$ ) matches it.
- **As a power amplifier** (80 Ba, 72 Ma): break the ring, feed RF in at one cavity and take it out several cavities later. The phase shift per cavity sets the wave's speed along the anode,  $v_p = \omega p / \phi$  ( $p$  = cavity pitch), and the beam drift is matched to it. That is the crossed-field amplifier of §5.6.
- Write the sketch with the cavities marked  $+$  /  $-$  according to  $\phi$  and draw the spokes in the retarding cavities; that sketch carries most of the marks.
- **Uses**: radar transmitters, microwave ovens (2.45 GHz), industrial heating, linear accelerators.

### 5.6 Crossed-Field Amplifier

## Explain the working principle of a circular-beam type microwave cavity-controlled crossed-field amplifier 3+7 7

PIN 2/24 72 Ma, 82 Bh · the magnetron turned into an amplifier

- A **crossed-field amplifier (CFA)**, also amplitron or platinotron: magnetron geometry,  $E \perp B$ , but with an RF input and an RF output.
- **Circular-beam (re-entrant beam)**: the electron stream circulates round the cathode continuously, as in a magnetron.
- **Cavity-controlled (non-re-entrant RF circuit)**: the anode cavities form a slow-wave line from input to output; between output and input sits an **RF absorber / insulator**, so the wave cannot circulate and the tube does not oscillate.
- **Working**:
  - The input wave travels along the anode at  $v_p = \omega p / \phi$ .
  - Electrons drift at  $E/B$ ; matched to  $v_p$  they **bunch into spokes** in the retarding phase of the wave (§5.5).
  - Spokes move with the wave and give up energy cavity by cavity  $\rightarrow$  the wave grows  $\rightarrow$  amplified output.
  - Left-over electrons carry little RF past the absorber and re-enter the input region.
- **Performance**: high efficiency (up to about 70%), broad band (about 10%), **low gain** (10 to 15 dB), high power.
- **Use**: final power stage of radar transmitters, driven by a TWT.



Circular CFA: input, output, absorber, spiralling electron flow.  
Student presentation 2078.

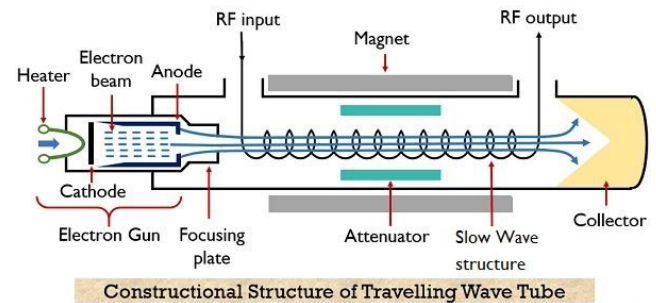
## 5.7 Travelling Wave Tube

### Describe the construction and operation of a travelling wave tube (TWT) 2+6

PIN 2/24 79 Bh, 80 Bh

#### Construction

- **Electron gun:** cathode, focusing plate, anode.
- **Helix:** a wire coil round the beam, the **slow-wave structure**; RF in at the gun end, out at the collector end.
- **Attenuator** midway along the helix: absorbs reflected waves so the tube cannot oscillate.
- **Focusing magnets** (solenoid or periodic permanent magnets) along the tube.
- **Collector:** collects the spent beam.
- **No resonant cavities** → very wide bandwidth.



TWT construction. Student presentation 2078.

#### Operation

1. RF on the helix travels round the turns at about  $c$ , but **along the axis** only at  $v_p \approx c \frac{\text{pitch}}{2\pi r}$  (a few percent of  $c$ ).
  2. The beam voltage is set so the electrons move slightly **faster** than  $v_p$ .
  3. The axial RF field velocity-modulates the beam → bunches form, and because beam and wave travel together the interaction lasts the **whole length**.
  4. Being slightly faster, the bunches sit in the retarding field and give energy to the wave, which grows exponentially → gain 30 to 60 dB.
- **Correction to the source** (PDF p38): the beam does **not** move at the velocity of light; the whole point of the helix is to slow the wave down to the beam. Example (tubes.py): pitch 1 mm, radius 2 mm gives  $v_p = 0.079c$ , matched by a beam of about 1.6 kV; a 1 kV beam moves at  $0.063c$ . verified/added
  - **Performance:** 0.5 to 95 GHz, octave bandwidth, 5 to 20% efficiency, mW to 10 MW pulsed. **Uses:** satellite transponders (long life), broadband receivers, radar, troposcatter links, EW jammers.

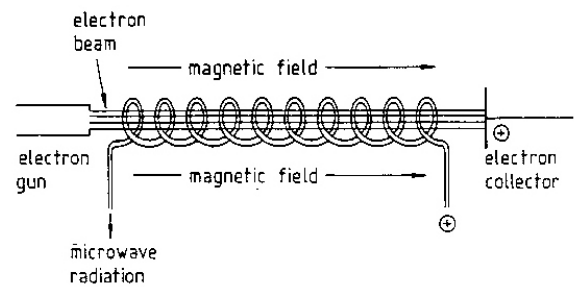
|                  | Two-cavity klystron         | TWT                             |
|------------------|-----------------------------|---------------------------------|
| Interaction      | at two gaps only            | continuous, whole helix         |
| RF circuit       | resonant cavities           | non-resonant slow-wave helix    |
| Bandwidth        | narrow (a few %)            | wide (an octave)                |
| Wave             | standing wave in cavities   | travelling wave                 |
| Oscillation risk | needs feedback to oscillate | attenuator needed to prevent it |

## 5.8 Backward Wave Oscillator and Slow-Wave Structures

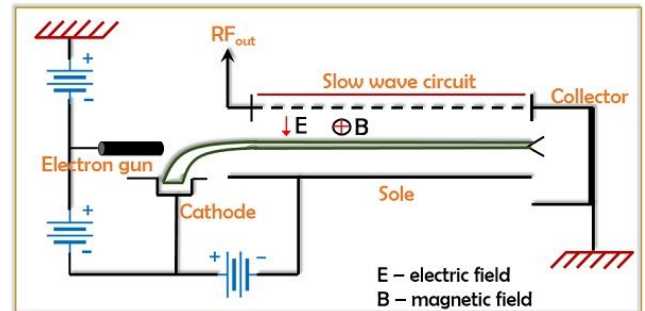
### What is bunching effect? Explain the working principle of a BWO with neat diagrams 5 2+8

**HOT 3/24** 71 Bh, 73 Ma, 74 Bh · 71 Bh and 74 Bh as short notes

- **BWO** (carcinotron): a TWT-family tube used as an **oscillator**, microwave up to terahertz.
- Electron gun, slow-wave structure (helix or folded line), collector, magnetic focusing. **No RF input**.
- The beam interacts with a **backward space harmonic**: its phase velocity is along the beam (so they stay in step) but its **group velocity**, the direction energy flows, is **towards the gun**.
- Bunches give energy to this wave; the energy travels back to the gun end, where it velocity-modulates newly arriving electrons: **built-in feedback** → self-sustained oscillation.
- **Output is taken at the gun end**; the collector end is matched.
- **Frequency is tuned electronically** by the beam voltage (it sets which backward harmonic is in step): very wide tuning range.
- M-type BWO: crossed fields with a negative **sole** electrode (lower figure).
- Uses: swept signal generators, THz spectroscopy and imaging, jammers.
- **Not** a klystron: some student notes describe buncher and catcher cavities and a feedback conductor; a BWO has none.



O-type BWO: output near the gun. Deck slide 40.



Schematic representation of Backward Wave Oscillator

Electronics Desk

M-type BWO with sole. Student presentation 2078.

### What do you mean by a slow (backward) wave structure? 2

73 Bh

- A **slow-wave structure** is a periodic RF circuit that makes the wave's axial phase velocity much smaller than  $c$ , so an electron beam of practical voltage can keep in step with it: helix, folded-back line, ring-bar, interdigital line, coupled cavities.
- It is **backward** when the space harmonic used has phase and group velocity in **opposite** directions: energy flows against the beam, as the BWO needs.

## 5.9 Low-Noise Amplifier

### Explain the construction and working principle of an LNA 2+6

**PIN 1/24** 73 Bh · **verified/added** no source note covers it; amplifier design in Ch 6

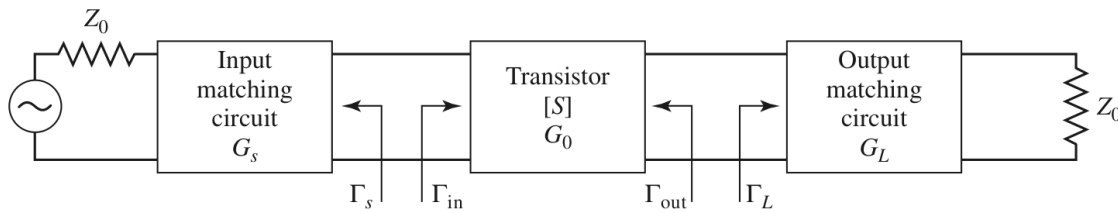
- **Purpose**: the first amplifier after the antenna. Its noise figure sets the whole receiver's, by Friis:

$$F = F_1 + \frac{F_2 - 1}{G_1} + \frac{F_3 - 1}{G_1 G_2} \quad (\text{one more term per stage})$$

e.g. a 1 dB, 20 dB-gain LNA before an 8 dB mixer gives 1.18 dB overall; the mixer first would give 8.02 dB (tubes.py).

- **Construction**:
  - **Low-noise device**: GaAs FET / HEMT (historically a parametric amplifier, maser or low-noise TWT).
  - **Input matching network**: presents the source reflection  $\Gamma_{opt}$  for **minimum noise figure**, not conjugate match.
  - **Output matching network**: conjugate match for gain.
  - **Bias network** with RF chokes and bypass capacitors; stabilising resistors so  $K > 1$ .
- **Working**: the signal is amplified with gain 10 to 20 dB while adding as little noise as possible (NF 0.5 to 2 dB); trade-off between minimum NF and maximum gain drawn as noise circles and gain circles on the Smith chart (Ch 6).

## The construction, drawn



Source, **input matching network** ( $\Gamma_s$ , set to  $\Gamma_{opt}$  for an LNA), transistor  $[S]$ , **output matching network** ( $\Gamma_L$ , conjugate for gain), load. Every amplifier in Ch 6 is this block diagram. Pozar Fig 12.2.

## 5.10 PYQ Mapping

### Theory questions, covered in this document

- **Limitations of conventional tubes; transit-time effect** → §5.1. 70 Ma 2, 77 Ch 2, 79 Bh 2
- **Bunching effect / density modulation** → §5.2. **Most repeated ask of the chapter.** 70 Bh 2, 71 Bh 2, 72 Ash 2, 73 Ma 2, 80 Ba 2, 80 Bh 2, 81 Bh 2
- **Velocity modulation and bunching in a (multicavity) reflex klystron** → §5.2 and §5.4. 72 Ma 5, 75 Bh 10
- **Two-cavity klystron amplifier** → §5.3. 69 Bh 5, 70 Ma 8
- **Multicavity klystron amplifier / oscillator; klystron oscillator** → §5.3. 71 Bh 8, 72 Ash 7, 78 Ch 8, 80 Ch 5, 82 Ba 4
- **Cross-field effect** → §5.5. 72 Ma 3, 78 Ch 2, 81 Bh 2
- **Cavity magnetron; oscillator using bunching and field effects** → §5.5. 70 Bh 8, 71 Ma 10, 79 Ch 5, 81 Bh 4
- **Magnetron with 45 / 60 / 90 degree phase shift** → §5.5. 74 Ma 8, 77 Ch 6, 80 Ba 6, 81 Ba 8, 72 Ma 7
- **Circular-beam cavity-controlled crossed-field amplifier** → §5.6. 82 Bh 7
- **TWT construction and operation** → §5.7. 79 Bh 6, 80 Bh 6
- **Backward wave oscillator** → §5.8. 71 Bh 5, 73 Ma 8, 74 Bh 5
- **Slow backward wave structure; LNA** → §5.8 and §5.9. 73 Bh 2+6

### Numerical content

- **This chapter has no numerical component:** no paper in 24 sets a tube calculation. The formulas above (bunching parameter, Hull cut-off, reflex transit condition, helix velocity) are there to strengthen descriptive answers, and each is checked in `src/tubes.py`.

### Corrections, and fixes made to the PYQ documents while building this chapter

- Source PDF p23: a klystron catcher converts only part of the beam energy (58% theoretical maximum), not all of it.
- Source PDF p38: a TWT beam moves far slower than light.
- Deck slide 27: reflex efficiency 22.78% is  $X'$  rounded; exact 22.7%.
- Student BWO slides describe klystron cavities; ignored.
- Sorted PYQ documents: 71 Bh's BWO short note was missing; 73 Ma is "bunching + BWO" and 73 Bh "slow backward wave structure + LNA"; 70 Ma asks no bunching question.

## Abbreviations used in these notes

Only these are ever shortened. Anything not on this list is written out in full.

| Short     | Means                                                       | Short    | Means                                                   |
|-----------|-------------------------------------------------------------|----------|---------------------------------------------------------|
| RF        | radio frequency                                             | MW, M/W  | microwave                                               |
| EM, EMR   | electromagnetic (radiation)                                 | TEM      | transverse electromagnetic mode                         |
| TE        | transverse electric mode                                    | TM       | transverse magnetic mode                                |
| WG        | waveguide                                                   | TL       | transmission line                                       |
| VSWR      | voltage standing wave ratio                                 | SWR      | standing wave ratio                                     |
| VNA       | vector network analyzer                                     | SA       | spectrum analyzer                                       |
| DUT       | device under test                                           | S-params | scattering parameters                                   |
| RL        | return loss                                                 | IL       | insertion loss                                          |
| ML        | mismatch loss                                               | NF       | noise figure                                            |
| BW        | bandwidth                                                   | SNR      | signal to noise ratio                                   |
| Tx, TX    | transmitter                                                 | Rx, RX   | receiver                                                |
| BTS       | base transceiver station                                    | GSM      | global system for mobile communications                 |
| EIRP      | effective isotropic radiated power                          | ERP      | effective radiated power                                |
| SAR       | specific absorption rate                                    | SARPs    | standards and recommended practices                     |
| HERP      | hazards of EM radiation to personnel                        | RADAR    | radio detection and ranging                             |
| ASR       | airport surveillance radar                                  | CW       | continuous wave                                         |
| Short     | Means                                                       | Short    | Means                                                   |
| TWT, TWTA | traveling wave tube (amplifier)                             | BWO      | backward wave oscillator                                |
| CFA       | cross-field amplifier                                       | LNA      | low noise amplifier                                     |
| HPA       | high power amplifier                                        | SWS      | slow wave structure                                     |
| O-type    | linear beam tube (axial $B$ -field)                         | M-type   | crossed-field tube ( $E \perp B$ )                      |
| MASER     | microwave amplification by stimulated emission of radiation | LASER    | light amplification by stimulated emission of radiation |
| MESFET    | metal-semiconductor FET                                     | HEMT     | high electron mobility transistor                       |
| GaAs      | gallium arsenide                                            | FET      | field effect transistor                                 |
| BJT       | bipolar junction transistor                                 | PIN      | p-type, intrinsic, n-type diode                         |
| IMPATT    | impact ionization avalanche transit-time                    | TRAPATT  | trapped plasma avalanche triggered transit              |
| LPF       | low pass filter                                             | HPF      | high pass filter                                        |
| BPF       | band pass filter                                            | BSF      | band stop filter                                        |
| MAG       | maximum available gain                                      | MSG      | maximum stable gain                                     |
| MIC       | microwave integrated circuit                                | MMIC     | monolithic microwave integrated circuit                 |
| ADS       | advanced design system                                      | IF       | intermediate frequency                                  |
| LO        | local oscillator                                            | RFIC     | radio frequency integrated circuit                      |
| PCB       | printed circuit board                                       | SMA      | subminiature version A connector                        |
| Short     | Means                                                       | Short    | Means                                                   |
| attr(s)   | attribute(s)                                                | no.      | number                                                  |
| freq      | frequency                                                   | info     | information                                             |
| ops       | operations                                                  | config   | configuration                                           |
| rm        | remove                                                      | vs, Vs   | versus                                                  |
| w/, w/o   | with, without                                               | Eg       | example / for example                                   |
| +Fig      | draw a figure                                               | (I)      | importance / need / significance                        |
| →         | leads to / therefore                                        | +        | and                                                     |
| =         | is / equals                                                 | norm     | normalized                                              |
| max, min  | maximum, minimum                                            | calc     | calculation / calculate                                 |
| eq, eqn   | equation                                                    | approx   | approximately                                           |
| pts       | points                                                      | coeff    | coefficient                                             |
| temp      | temperature                                                 | params   | parameters                                              |

**Standard and component terms** (proper names and symbols, kept as standard): **Magic Tee** 180° hybrid junction (E-plane & H-plane tee combined) · **Directional Coupler** 4-port forward/reverse coupled wave device · **Circulator** non-reciprocal multi-port ferrite device ( $1 \rightarrow 2 \rightarrow 3 \rightarrow 1$ ) · **Gunn Diode** transferred electron device (GaAs / InP, negative resistance) · **Bolometer** power detector (barretter: PTC, thermistor: NTC) · **Calorimeter** power measurement via heat transfer (static, circulating, dry, flow).

**Key mathematical symbols:**  $Z_0$  characteristic impedance ·  $Y_0$  characteristic admittance ·  $\Gamma$  reflection coefficient ·  $S$  scattering matrix ·  $K$  Rollett's stability factor ·  $\Delta$  determinant of S-matrix ·  $\mu$  single parameter stability metric ·  $G_T$  transducer gain ·  $G_{TU}$  unilateral transducer gain ·  $f_c$  cutoff frequency ·  $\lambda_g$  guide wavelength ·  $v_p$  phase velocity ·  $v_g$  group velocity.

**Exam-paper month codes** (used in the year tags): Ba = Baishakh · Bh = Bhadra · Ash = Ashwin · Ma = Magh · Ch = Chaitra.

**Bold** year tags are Regular exams, plain are Back exams.

Monospace year tags are BEI (EX 716, IV/I); normal font tags are BEX (EX 752 / EG 785EX, IV/II).

**Tier chips** over 24 papers, 2069–2082 BS: **TOP /24** = topic asked in 7 or more papers · **HOT /24** = 3 to 6 papers · **PIN /24** = 1 or 2 papers but worth  $\geq 6$  marks.

**Mark chips** such as **2+5** are the marks that question actually carried. **syllabus, no PYQ yet** marks a syllabus topic that has never been examined but is written up anyway. **verified/added** marks anything researched and added that was not in the source notes.